

(a) Paired observations under moment-designed profiles

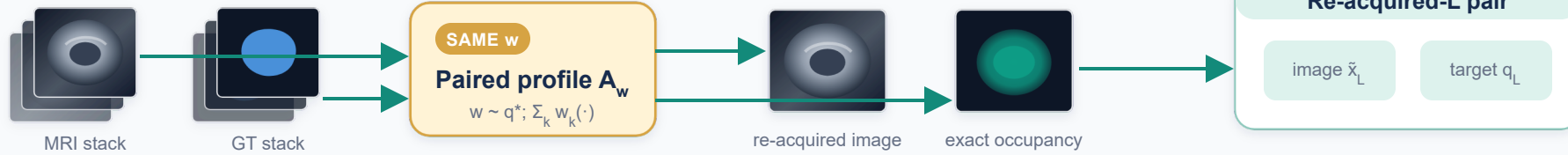
NATIVE LABEL

Identity path preserves native anatomy



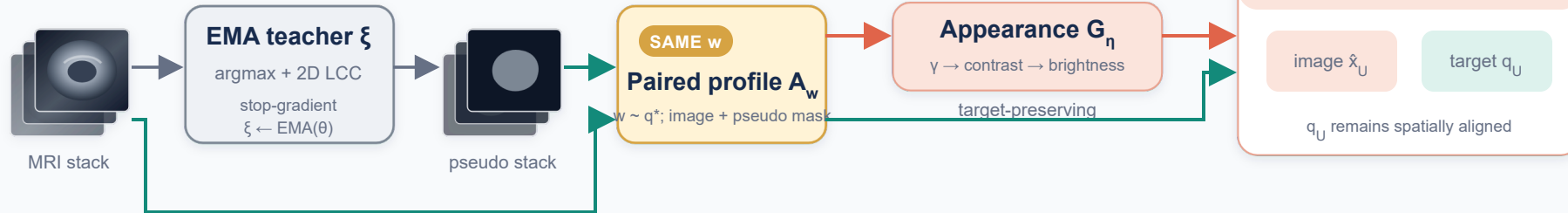
PAIRED LABELED

The same profile transforms image and exact mask

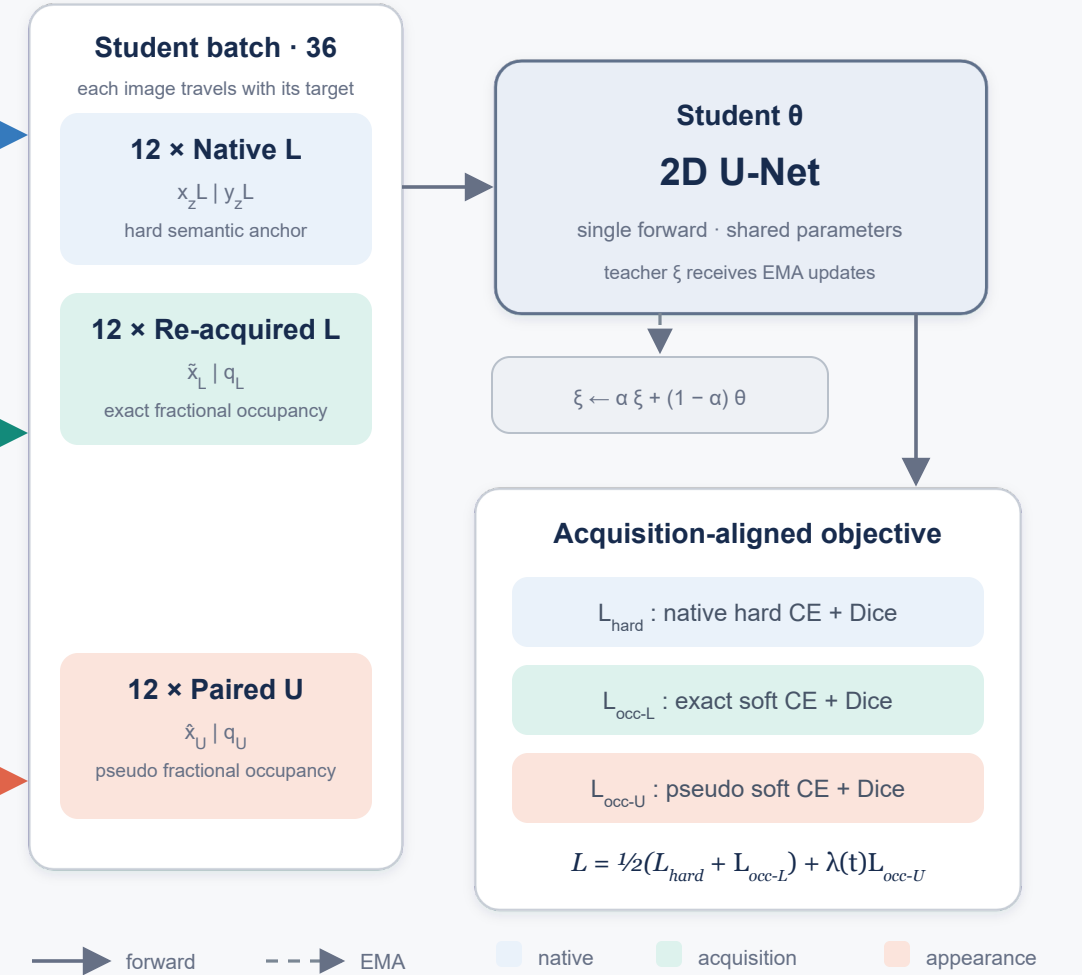


UNLABELED U

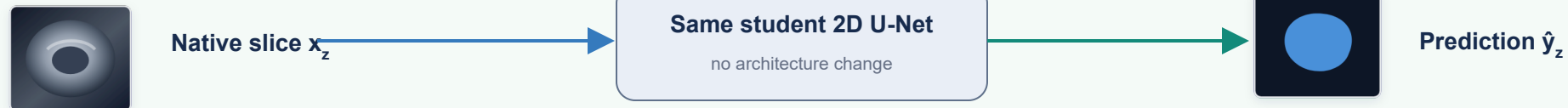
Acquisition first; target-preserving appearance second



(b) Shared optimization



(c) Inference $\cdot$ unchanged single-slice path



Training-only operations are removed

all training-only observations are removed
no extra parameters or inference latency